

PROBLEMA I

①

a)
(2 puntos)

$$G(s) = \frac{100}{s(s-1)}$$

$$s = -2 \pm j2 \rightarrow \omega_n = \sqrt{2^2 + 2^2} \rightarrow T \leq \frac{2\pi}{40\omega_n} = 0.055$$

$$T = 0.05 \text{ segundos}$$

(0.5 p)

$$G(z) = Z \left[\frac{1-e^{-sT}}{s} \frac{100}{s(s-1)} \right] =$$

$$= (1-z^{-1}) Z \left[\frac{100}{s^2(s-1)} \right] = 100 \frac{z-1}{z} Z \left[\frac{1}{s^2(s-1)} \right] =$$

$$= 100 \frac{z-1}{z} Z \left[\frac{A}{s^2} + \frac{B}{s} + \frac{C}{s-1} \right] =$$

(0.2 p)

$$A = \frac{1}{s^2(s-1)} \Big|_{s=0} = -1$$

$$B = \frac{d}{ds} \left[\frac{1}{s-1} \right] \Big|_{s=0} = \frac{-1}{(s-1)^2} \Big|_{s=0} = -1$$

(0.2 p)

$$C = \frac{1}{s^2(s-1)} \Big|_{s=1} = 1$$

(0.3 p) (0.3 p) (0.3 p)

$$= 100 \frac{z-1}{z} Z \left[\frac{-1}{s^2} + \frac{-1}{s} + \frac{1}{s-1} \right] = 100 \frac{z-1}{z} \left[-\frac{0.05z}{(z-1)^2} + \frac{-z}{z-1} + \frac{z}{z-e^{0.05}} \right]$$

$$= 100 \frac{z-1}{z} \left[\frac{-0.05z/(z-1.0513) - z/(z-1)(z-1.0513) + z/(z-1)^2}{(z-1)^2(z-1.0513)} \right]$$

$$= 100 \left(\frac{+0.05 \cdot 1.0513 - 0.05z - z^2 + 1.0513z + z - 1.0513 + z^2 - 2z + 1}{(z-1)(z-1.0513)} \right) =$$

$$= 100 \left(\frac{0.0013z + 0.0013}{(z-1)(z-1.0513)} \right) = 100 \cdot 0.0013 \frac{(z+1)}{(z-1)(z-1.0513)} =$$

$$= \boxed{0.13 \frac{(z+1)}{(z-1)(z-1.0513)}} \leftarrow (0.2 p)$$

b) $G(z) = 0.13 \frac{(z+1)}{(z-1)(z-1.0513)} = \frac{Y(z)}{u(z)}$ (1.5 points) 5 $\frac{z}{R}$ (0.2 p)

$$Y(z) = 0.65 \frac{(z+1)}{(z-1)(z-1.0513)}$$

$$\frac{Y(z)}{z} = 0.65 \frac{(z+1)}{z(z-1)(z-1.0513)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z-1.0513} \quad \leftarrow (0.2 p)$$

$$\begin{cases} A = 0.65 \frac{(z+1)}{z(z-1)(z-1.0513)} \Big|_{z=0} = \frac{0.65}{1.0513} = 0.6183 \\ B = 0.65 \frac{(z+1)}{z(z-1)(z-1.0513)} \Big|_{z=1} = \frac{1.3}{-0.0513} = -25.3411 \\ C = 0.65 \frac{(z+1)(z-1.0513)}{z(z-1)(z-1.0513)} \Big|_{z=1.0513} = \frac{1.33}{-0.0519} = -24.7371 \end{cases} \quad \leftarrow (0.2 p)$$

$$Y(z) = \frac{0.6183 z}{z} - 25.3411 \frac{z}{z-1} + 24.7371 \frac{z}{z-1.0513}$$

$$\downarrow z^{-1} \quad \downarrow (0.3 p) \quad \downarrow (0.3 p) \quad \downarrow (0.3 p)$$

$$\boxed{\{Y(k)\} = 0.6183 \{ \delta_0(k) \} - 25.3411 \{ 1^k \} + 24.7371 \{ 1.0513^k \}}$$

c) $G(z) = 0.13 \frac{(z^{-1} + z^{-2})}{(1-z^{-1})(1-1.0513z^{-1})} = \frac{Y(z)}{u(z)} \quad \leftarrow (0.2 p)$ (1 point)

$$Y(z) [1 - 2.0513z^{-1} + 1.0513z^{-2}] = u(z) [z^{-1} + z^{-2}] 0.13$$

$$z^{-1} \downarrow$$

$$Y(k) = 0.13 u(k-1) + 0.13 u(k-2) + 2.0513 y(k-1) - 1.0513 y(k-2) \quad \leftarrow (0.4 p)$$

Si $\{u(k)\} = \{5, 0, 0, 0, \dots\}$:

$$k=0 \rightarrow y(0) = 0$$

$$k=1 \rightarrow y(1) = 0.13u(0) = 5 \cdot 0.13 = 0.65$$

$$k=2 \rightarrow y(2) = 0.13u(1) + 2.0513y(1) = 1.9833$$

$\begin{matrix} 0.13 \cdot 5 & 0.65 \end{matrix}$

$$k=3 \rightarrow y(3) = 2.0513y(2) - 1.0513y(1) = 3.3850$$

$\begin{matrix} 1.9833 & 0.65 \end{matrix}$

$\{y(k)\} = \{0, 0.65, 1.9833, 3.3850, \dots\}$

← (0.4p)

d) (0.5 puntos) DADO QUE EL SISTEMA PRESENTA UN POLO INESTABLE EN $z=1.0513$ LA SALIDA DIVERGE (NO TIENE A NINGÚN VALOR FINAL DE CONVERGENCIA).